# Mauro Faccin
Curriculum Vitæ

Dip. Fisica e Astronomia
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Preferred pronouns: he/him


## Actual Position

2024-today **DFA at University of Padova** — University of Padova, Italy
Interests/Research line
- Analysis of climate data and their interaction with human made networks
- Mesoscopic structures and their influence on the dynamical system.

## Professional Experience

2023 - 2024 **University of Bologna, (DIMEC and INFN)** — Bologna, Italy
Postdoctoral researcher
- Analysis of medical data through the use of Machine learning and complex networks combined approaches.
- Prediction of disease survival
- Analysis of information dynamics in social networks and medical systems.

2022 - 2023 **Istituto Professionale Statale "G. B. Garbin"** — Schio, Italy
High school math teacher

2021 - 2022 **Université de Paris, (CEPED and IRD)** — Paris, France
Postdoctoral researcher
- Vaccines hesitancy unveiled through hypergraph analysis.
- We analyse the discussion on microblogging services as a dynamical system spreading on a hypergraph.
- Analysis of the textual content and its dynamical and geographic properties.

2020 - 2021 **University of Turin (Computer Science)** — Torino (Italy)
Visiting

2015 - 2020 **Université Catholique de Louvain (ICTEAM)** — Louvain la Neuve, Belgium
MOVE IN Louvain Research Fellow
- Interplay between complex networks and dynamical systems.
- Making use of mathematical tools from information theory we could analyse dynamical systems, their evolution and the emerging mesoscopic structures.
- We successfully built a bridge between statistical inference and dynamical approaches in community or block detection.
- Development of a `python` library.
- Spatial analysis of disease spreading with compartmental models. (tuberculosis incidence inference in African countries)

2011 - 2015 **ISI Fundation** — Torino, Italy
Junior researcher
- Quantum meets complex networks.
- I contributed to a multidisciplinary team interested in using tools from complex networks to study quantum system.

2006 - 2011 **Bifi Institute** — Zaragoza, Spain
PhD
- Use of Ising-like models to characterize biopolymers with application to protein folding and protein sequencing.
- Development and release of corresponding software

## Fellowships and Projects

2021 - 2022 TRACTRUST (ANR-20-COVI-0102)
2017 - 2021 Innoviris Iris-TB (D1.31402.007-F)

2015 - 2017 **MOVE-IN Louvain Fellowship (with support of Marie Curie Actions)**

## Program Committees

| | | |
|---|---|---|
| 1 Dec 2021 | **Complex Networks 2021** | Madrid, Spain |
| 1 Dec 2020 | **Complex Networks 2020** | Madrid, Spain |
| 10 Dec 2019 | **Complex Networks 2019** | Lisbon, Portugal |
| 22 Feb 2019 | BeNet 2019 | Hasselt, Belgium |
| 11 Dec 2018 | **Complex Networks 2018** | Cambridge, UK |
| 29 Nov 2017 | **Complex Networks 2017** | Lyon, France |
| 30 Nov 2016 | **Complex Networks 2016** | Milan, Italy |

## Organizing Committees

25 Oct 2024 **Networks Days** Padova, Italy
Local committee

August 2019 **Network Reconstruction Science-athon** Beaumont, Belgium
Coorganized with Alexandre Bovet, Benjamin Chiem, Leo Gutierrez, Alexey Medvedev, Leto Peel, Mridul Seth

15 Dec 2016 **Belgian Network Research Meeting (BeNet) 2016** Louvain-la-Neuve, Belgium
Coorganized with Sophie Béreau, Jean-Charles Delvenne, Yerali Gandica, Leo Gutierrez, Leto Peel, Michael Schaub

21 Sep 2016 **Conference on Complex Systems 2016 workshop on Coarse-graining Complex Systems** CCS2016, Amsterdam, Netherlands
Coorganized with Leto Peel, Renaud Lambiotte, Michael Schaub

2 Jun 2014 **Quantum Frontiers for Network Science** NetSci 2014, Berkeley, California
Coorganized with Filippo Radicchi and Zoltan Zimborás

## Reviewer

- Advances in Complex Systems
  World Scientific
- Applied Network Science
  Springer Nature
- Entropy
  Multidisciplinary Digital Publishing Institute (MDPI)
- Journal of Complex Networks
  Oxford University Press
- Journal of Statistical Mechanics
  IOP
- Natural Computing
  Springer Netherlands
- PLOS ONE
  Public Library of Science (PLoS)
- PRA, PRB, PRE, PRL, PRX, PRResearch
  American Physical Society
- Science Advances
  American Association for the Advancement of Science
- Scientific Reports
  Nature Portfolio

## Teaching

2023-2024 **DAT301 Social Network Analysis** University Mohammed VI Polytechnic
with F. Gargiulo and A. Apolloni

2015-2020 **LINMA2111 Algorithms and Complexity** Université Catholique de Louvain
with J-C Delvenne

2005 **Physics and Math at primary school** Hetauda, Nepal
NEST
2004 **guide in Science Museum** Padova, Italy
Organizer: INFN

## STUDENT ADVISING

2024 N. Barbieri University of Bologna
**Corelatore** tesi magistrale with F. Durazzi and D. Remondini in Applied Physics
Integration of artificial intelligence and network approaches to classify data in social networks

2019 G. Izzi University of Turin
Tesi magistrale with A. Semeraro and G. Ruffo (informal tutor). Computer Science
Analysis of temporal patterns and forecasting of popularity for programming languages on Stack Overflow platform

2016 H. Devillez UCLouvain
Applied Maths (project with L. Peel)

2016 M. Huet UCLouvain
Elec. Eng. (project with L. Peel)

## EDUCATION

2020 **24CFU** Univestità telematica Pegaso
Discipline antropo-psico-pedagogiche e in metodologie e tecnologie didattiche

14 Sept 2012 PhD in Physics University of Zaragoza
supervisor: prof. Pierpaolo Bruscolini
Analysis and Development of Algorithms for the Characterization of Biopolymers and Application to Protein Sequencing and Protein Folding

15 Oct 2008 **Master in Physics and Technology** University of Zaragoza
supervisor: prof. Pierpaolo Bruscolini
Coevolution of Strategy and Population Structure in the Prisoner Dilemma

10 Oct 2005 **MSc in Physics** University of Padova
supervisor: prof. Enzo Orlandini
Beris Edwards Model Simulations for Liquid Crystals in Oscillatory Poiseuille Flux

## MAINTEINED PROJECTS

- AISA
  Python module aimed at partitioning a network through Autoinformation maximization
- CartoTB
  Cartography of predicted tuberculosis in high burden settings

## PREPRINTS

- Temporal and geographic analysis of the Hydroxychloroquine controversy in the French Twittosphere arXiv:2304.14075
  M. Faccin, E. Schultz and F. Gargiulo

## PUBLICATIONS

- Systemic collapse emergence from climate-induced perturbations in transportation networks Proc. Natl. Acad. Sci. U.S.A. 123 (42) e2532180123, (2026)
  M. Faccin, T. Scagliarini, M. De Domenico
- Leveraging complex network features improves vaccine stance classification Scientific Reports, 15(1) (2025)
  F. Durazzi, N. Barbieri, M. Faccin, D. Gori, D. Remondini

- Analyzing Satellite Imagery to Target Tuberculosis Control Interventions in Densely Urbanized Areas of Kigali, Rwanda: Cross-Sectional Pilot Study
  JMIR Public Health Surveill 2025;11:e68355
  M. Faccin, C. Geenen, M. Happaerts, S. Ombelet, P. Migambi, E. André
- Mapping of regions with low tuberculosis notification and estimation of diagnostic gaps in Cameroon evidence from OpenStreetMap and WorldPop data
  Scientific Reports, 15, Article number: 4992 (2025)
  C.G. Jouego, T.N. Collins, B. Tegomoh, C. Geenen, P.M. Netongo, E. André, M. Faccin
- Measuring dynamical systems on directed hypergraphs Phys Rev E, 106 (034306), 2022
  M. Faccin
- Assessing the influence of French vaccine critics during the two first years of the COVID-19 pandemic PLOS ONE, 17(8): e0271157
  M. Faccin, F. Gargiulo, L. Atlani-Duault, J.K. Ward
- Data-driven identification of communities with high levels of tuberculosis infection in the Democratic Republic of Congo. Sci Rep 12, 3912 (2022)
  M. Faccin, O. Rusumba, A. Ushindi, M. Riziki, T. Habiragi, F. Boutachkourt, E. André
- State aggregations in Markov chains and block models of networks
  Phys. Rev. Lett., 127, 078301 (2021)
  M. Faccin, M.T. Schaub and j.-C. Delvenne
  Figure selected for the journal cover
- Complex Networks from Classical to Quantum Communications Physics 2-53 (2019)
  J. Biamonte, M. Faccin, M. De Domenico
- Outbreak of multidrug-resistant tuberculosis in South Africa undetected by WHO-endorsed commercial tests: an observational study
  The Lancet Infectious Diseases 2018
  Makhado, Matabane, Faccin, Pinçon, Jouet, Boutachkourt, Goeminne, Gaudin, Maphalala, Beckert, Niemann, Delvenne, Delmée, Razwiedani, Nchabeleng, Supply, de Jong, André
- Entrograms and coarse graining of dynamics on complex networks
  Journal of Complex Networks
  M. Faccin, M.T. Schaub, J-C. Delvenne
- Chiral Quantum Walks Phys. Rev. A 93, 042302
  D. L. Lu, J. D. Biamonte, J. Li, H. Li, T. H. Johnson, V. Bergholm, M. Faccin, Z. Zimborás, R. Laflamme, J. Baugh, and S. Lloyd
- Mapping the topography of a protein energy landscape
  J. Am. Chem. Soc., 2015, 137 (46), pp 14610–14625
  R. D. Hutton, J. Wilkinson, M. Faccin, E. M. Sivertsson, A. Pelizzola, A. R. Lowe, P. Bruscolini, L. S. Itzhaki
- Community Detection in Quantum Complex Networks
  Phys. Rev. X 4 (4 2014), p. 041012
  M. Faccin, P. Migdał, T. H. Johnson, V. Bergholm, and J. D. Biamonte
- Degree Distribution in Quantum Walks on Complex Networks
  Phys. Rev. X 3 (4 2013), p. 041007
  M. Faccin, T. Johnson, J. Biamonte, S. Kais, and P. Migdał
- Quantum Transport Enhancement by Time-Reversal Symmetry Breaking
  Sci. Rep. 3 (2013). Ed. by NPG, p. 2361
  Z. Zimboras, M. Faccin, Z. Kadar, J. Whitfield, B. Lanyon, and J. Biamonte
- MS/MS Spectra Interpretation as a Statistical-Mechanics Problem
  Analytical Chemistry 85.10 (2013), pp. 4884–4892
  M. Faccin and P. Bruscolini
- Ground-state spin logic Europhysics Letters 99.5 (2012), p. 57004
  J. D. Whitfield, M. Faccin, and J. D. Biamonte
- Analysis of the equilibrium and kinetics of the ankyrin repeat protein myotrophin
  The Journal of Chemical Physics 134.7, 075102 (2011), p. 075102
  M. Faccin, P. Bruscolini, and A. Pelizzola

## Conferences (Oral Contributions)

2 Sept 2025 Climate-induced delay propagation in rail networks as a non-Markovian process
CCS 2025
M. Faccin, T. Scagliarini, M. De Domenico

22 Jul 2025 **Climate-induced delay propagation in rail networks as a non-Markovian process**
StatPhys29 (Complex Networks)
M. Faccin, T. Scagliarini, M. De Domenico

10 Oct 2023 **International topic distribution in the HCQ controversy on Twitter**
CCS/Italy @ Napoli (Italy)
M. Faccin, E. Schultz, F. Gargiulo

7 Mar 2023 **Dynamical systems as a mean for complex systems analysis**
seminar at the University of Bologna
M. Faccin

9 Nov 2022 **Data-driven mapping of tuberculosis incidence allows identifying communities with high circulation of tuberculosis in the Democratic Republic of Congo**
The Union Conference 2022 – Online
M. Faccin, C. Nsop Mulaj, E. Swana Kawana, T. Yav Ikum, E. Andé

9 Nov 2022 **Identification of urban areas presenting high incidence of tuberculosis using satellite images allows increasing the yield of active case finding in Kigali, Rwanda.**
The Union Conference 2022 – Online
M. Faccin, P Uwizeye, F. Nsabimana, P. Migambi, E. André

17 Oct 2022 **Use of compartmental models to detect high TB incidence rates in South-Kivu (DRC)**
CCS 2022 @ Palma de Mallorca/Online
M. Faccin, E. André

20 June 2022 **Reach of vaccine-critical content on Twitter during the COVID-19 pandemic**
French Regional Conf. on Complex Systems (FRCCS) @ Paris
M. Faccin, F. Gargiulo
Given by F. Gargiulo

30 May 2022 **Outreach of vaccine-critic content on French-speaking Twitter**
Complenet 2022 – Online
M. Faccin, F. Gargiulo, J. Ward

2 Dec 2021 **Autoinformation state aggregation** Complex Networks 2021 @ Madrid (Spain)
M. Faccin, M. T. Schaub, J-C. Delvenne

30 Nov 2021 **Vaccine-critical discourse on French-speaking Twitter and its outreach**
Complex Networks 2021 @ Madrid (Spain)
M. Faccin, F. Gargiulo, J. Ward
Best lightning talk award

19 Oct 2021 **State aggregation for dynamical systems: An information-theoretic approach**
Math Dept, Oxford University, (UK – Online)
Network Seminar 2021

7-9 Dec 2020 **Using predictive models and risk assessment tools to find unreported TB cases**
Washingthon DC, USA (online)
Global Digital Health Forum 2020

20-24 Oct 2020 **Data-driven TB active-case finding in South Kivu (DRC)** Paris, France (online)
The Union 2020

10-12 Dec 2019 **Auto-information in non-Markovian diffusion systems** Lisbon, Portugal
Complex Networks 2019

22 Feb. 2019 **Variable aggregation for dynamical systems on networks** Hasselt, Belgium
Benet 2019

23-28 Sept. 2018 **A dynamical approach to block structure analysis** Thessaloniki, Greece
CCS 2018

11-15 Jun 2018 **Dynamical Markov Modules** Paris, France
Netsci 2018

21-24 Mar 2017 **Community detection in networks with non-Markovian dynamics** Dubrovnik, Croatia
CompleNet 2017

30 Nov-2 Dec 2016 **Behind Communities: Dynamics Define Modules** Milan, Italy
ComplexNetworks 2016

19-22 Sep 2016 **Beyond Communities: Dynamical Markov Modules** Amsterdam, Netherlands
CCS 2016

May 30-June 3 2016 **Mesoscopic Structure And Diffusion Process Memory** Seoul, Korea
NetSci 2016

10-15 Jan 2015 **Mesoscopic Structures and Diffusion Process Memory** Wrocław, Poland
NetSci-X 2016

2-6 Jun 2015 **Structural controllability of EEG temporal causal networks** Zaragoza, Spain
NetSci 2015

22-26 Sep 2014 **Quantum Approaches to Community Detection** Lucca, Italy
ECCS 2014

12-14 Mar 2014 **Community detection on quantum systems** Bologna, Italy
CompleNet 2014

22-24 Jan 2014 **Community Detection on Quantum Systems** Zaragoza, Spain
BIFI2014

24-26 May 2013 **Degree distribution in quantum complex networks** Waterloo, Canada
IQC workshop on quantum computation and complex networks

6-8 Nov 2012 **Ground state spin logic** Torino, Italy
Tensor Network States and Algebraic Geometry Workshop

2-4 Feb 2011 **Equilibrium and kinetics of the ankyrin repeat protein Myotrophin through a simple statiscical mechanical model** Zaragoza, Spain
V BIFI National Conference

5-7 Feb 2009 **Bayesian Characterization of Protein Tandem Mass Spectra** Zaragoza, Spain
IV BIFI National Conference

8-9 May 2008 **Noise description in MSMS mass spectrometry, looking for a de novo sequencing algorithm** San Sebastian, Spain
VI Jornadas Aragonesas de Estabilidad y Plegamiento de Proteínas

## Conferences (Poster Contributions)

1 Dec 2021 **Directed hyper-graphs and dynamical systems**
Complex Networks 2021 @ Madrid (Spain)
M. Faccin, F. Gargiulo

17-25 Sept 2020 **State aggregation in non-Markovian systems by auto-information maximization**
Rome, Italy
Netsci 2020

17-25 Sept 2020 **The variability of network structures inferred from time series data** Rome, Italy
Netsci 2020

10-12 Dec 2019 **The variability of network structures inferred from time series data** Lisbon, Portugal
Complex Networks 2019

1-3 July 2019 **Variable aggregation on (broadcasting) networks** Trento, Italy
CCS/Italy 2019

23-25 Mar 2016 **Memory and Mesoscopic Structures in Diffusion Processes** Dijon, France
Complenet 2016

7-10 Jul 2010 **Folding Pathways Characterization for Ankyrin Repeat Protein Myotrophin**
Zaragoza, Spain
IV Spanish Portuguese Biophysical Congress

2-6 Jun 2008 **Statistical characterization of fragment ions in MSMS mass spectrometry**
Sitges, Spain
XXI Sitges Conference on Statistical Mechanics: Statistical Mechanics of Molecular Biophysics

## Computer skills

Coding **Python, C, Matlab, Fortran**
Shell **Bash, iPython/Jupyter**
Versioning **GIT, Subversion, Mercurial**
Plotting **Matplotlib (python), Gnuplot, PGFPlots (LaTeX)**
Typesetting **LaTeX, LibreOffice, MSOffice, Markdown**
Imaging **Gimp (Raster), Tikz (LaTeX), Inkscape (Vectorial), Rawtherapee (RAW)**
Web **HTML5, CSS/SCSS, static site generators, Wordpress**

## Languages

Italian **Mother tongue**
English **Fluent**
Spanish **Fluent**
French **Fair**

Mauro Faccin